Fiducials & Tooling Holes

The alignment infrastructure SMT machines need

Two small features without which assembly is impossible

Audience: PCB Designers

Vision alignment • Fiducial types • Tool holes • IPC rules • Placement

Why Fiducials + Tool Holes Matter

Vision alignment

P&P machine cameras find fiducials → know where every pad is.

Mechanical alignment

Tool pins through holes lock panel in fixture.

Rotation correction

Fiducial pair detects panel rotation/skew.

Per-board fiducials

Each board has its own (in addition to panel ones).

BGA fiducials

Local fiducials around fine-pitch BGA placement.

Without them

SMT line CANNOT process the board automatically.

Fiducial Types

- Global fiducial: 3 per panel (top-left, top-right, bottom-left)
- Local fiducial: 2-3 around each fine-pitch IC (BGA, QFP-0.4mm)
- Standard size: 1 mm diameter solid copper dot
- Solder-mask opening: 2 mm dia (1 mm dot + 0.5 mm clearance ring)
- Plain copper surface — no plating finish (avoid ENIG glare)
- Color: copper with green/clear mask ring around
- Vision systems look for THE CONTRAST between copper and mask
- Position: not too close to board edge ($\geq$ 5 mm clearance)

Tooling Hole Specifications

- Diameter: 3.0 mm or 4.0 mm (CM-specific)
- Count: 4 minimum per panel (corners of rail)
- Plated or unplated: CM preference (unplated more common)
- Position: ~5 mm from panel edge, ~5 mm spacing from each other
- Asymmetric placement: shift one inward → prevents panel flipping
- Position tolerance: ±0.05 mm (very tight)
- Don't route copper near tooling holes — clearance ≥ 1 mm
- Specify on Edge.Cuts or dedicated 'Tooling' layer in EDA tool

Where to Place Fiducials

- Panel fiducials: 3 on the rail — top-left, top-right, bottom-left
- Asymmetric pattern (3 corners, not 4) tells vision which way the panel sits
- Board fiducials (in addition to panel): 2-3 on each individual board
- Local fiducials for fine-pitch ICs: 2 fiducials, opposite corners of IC, within 5 mm
- Avoid placing fiducial near a connector or pad (visual confusion)
- Don't put fiducials inside a copper pour — kills the contrast
- Mask opening MUST be present — no mask = no contrast

Local Fiducials for BGAs

- Fine-pitch packages (BGA < 0.65 mm, QFP-0.4 mm) need LOCAL fiducials
- Why: even with panel fiducials, panel can warp/shift slightly during process
- Local fiducials let the vision system re-align before placing the specific IC
- 2 fiducials per IC, diagonal corners of the package footprint
- Placed in the keep-out region OUTSIDE the IC body
- Same 1 mm dia, 2 mm mask opening as standard fiducial
- Required by IPC-7351 for high-density layouts

Material + Finish

- Copper surface — NOT plated with HASL/ENIG/OSP
- Reasoning: matte copper has better camera contrast than shiny finishes
- Some fabs apply OSP to the fiducial — acceptable but not ideal
- ENIG fiducials reflect IR-light, confuse some vision systems
- Specify on fab quote: 'Fiducials: bare copper, no surface finish'
- Solder-mask opening AROUND the fiducial creates clean contrast boundary
- If your fab can't do bare-copper fiducials, OSP is the second-best choice

DRC & DFM Rules

- Fiducials need a copper-free zone around them (≥ 1 mm clearance)
- Soldermask opening ≥ 2 mm (1 mm fiducial + 0.5 mm ring on each side)
- Don't place fiducials in keepout zones (rails) — must be on a flat surface
- Tooling holes: drill hole with no copper pad
- Tooling hole keepout: ≥ 0.5 mm copper-free margin around the hole
- EDA tools have fiducial primitives — use them; don't draw manually
- KiCad: Footprints library → Fiducial:Fiducial_1mm_Mask2mm.kicad_mod

Common Mistakes

- Only 2 fiducials per panel (need 3) → vision can't compute rotation
- Symmetric panel without anti-rotation feature → loaded backward, all parts wrong
- Fiducials too close to a connector → vision confused
- Fiducials with full mask coverage (no opening) → no contrast, vision fails
- Fiducials in a copper pour → no contrast
- Missing local fiducials on BGAs > 200 balls → placement error
- Tooling holes too small (< 3 mm) — CM fixture pins don't fit
- Tooling holes plated when CM expected unplated — fixture pin damage

Cost Impact

- Fiducials + tooling holes are FREE — included in standard fab cost
- Time investment: ~5 min in EDA tool per panel design
- Cost of MISSING fiducials: CM rejects panel + 1-week redesign delay
- Cost of missing local fiducials on BGAs: high placement defect rate
- Bottom line: zero excuse to skip — pure design discipline

How to Add in KiCad

- Fiducial: Edit → Add Footprint → Fiducial:Fiducial_1mm_Mask2mm.kicad_mod
- Tooling hole: Edit → Add Footprint → MountingHole:MountingHole_3.2mm_M3 (unplated)
- Place panel fiducials on a 'Rails' / 'Edge.Cuts' layer
- Local fiducials: in the board footprint, near fine-pitch components
- DRC: ensure copper keepout around each fiducial + tool hole
- Document in fab quote: 'Bare copper fiducials, 3 per panel + 2 per BGA'

CM-Specific Variations

- JLCPCB: 3 fiducials on rails (top-L, top-R, bottom-L)
- PCBWay: similar; their preferred panel template online
- Foxconn / Jabil / Flex: custom rules per facility — ALWAYS ask before designing
- Indian CMs: standard IPC-A-610 + their fixture requirements
- When changing CM mid-product → re-verify panel + fiducial requirements
- Some CMs prefer fiducials on assembled side (top); others both sides

Key Takeaways

Key takeaways from this lecture

3 fiducials per panel

Asymmetric placement.

2 local per BGA

Fine-pitch alignment.

Tooling holes 3-4 mm

4 per panel, asymmetric.

Bare copper fiducials

No HASL/ENIG.

Mask opening ≥ 2 mm

Contrast for vision.

Free to add

No excuse to skip.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC-7351 — landpattern + fiducial guidance
- JLCPCB / PCBWay fiducial requirements
- KiCad fiducial footprints library
- Phil's Lab YouTube — panelization videos cover fiducials
- KiKit panelization (auto-generates fiducials)
- CM-specific design rule documents

Thank you • Build something this week.